ULTIMATE CHEMISTRY STRATEGIC ANALYSIS

Based on Sets 56/5/1, 56/7/1, and HFG1E/5 • Final Revision Masterfile

Created by Pragya for Thunderstudy

STEP 0 — INPUT CHECK

- **PDFs Uploaded:** 3 Question Papers found.
- **Years Detected:** 2023-2024 pattern (70 Marks, Sections A-E).
- **Syllabus Match:** ✅ **PERFECT MATCH.** New Reduced Syllabus (2024 Onwards) detected. (No p-block, Solids, or Polymers).

STEP 1 — SYLLABUS FILTERING

A) VALID QUESTIONS: All questions in the uploaded PDFs are valid for the current session.

B) REMOVED TOPICS: Zero detected. All chapters covered are relevant.

STEP 2 — CHAPTER WEIGHTAGE

Chapter Name	Importance Level	Avg. Qs per Paper
Electrochemistry	HIGH (Critical)	4-5 Questions
Chemical Kinetics	HIGH (Critical)	4-5 Questions
Aldehydes, Ketones & Acids	HIGH (Critical)	3-4 Questions
Solutions	HIGH	3-4 Questions
Coordination Compounds	Medium	3 Questions
d- and f-Block Elements	Medium	3-4 Questions
Amines	Medium	3 Questions

Chapter Name	Importance Level	Avg. Qs per Paper
Haloalkanes & Haloarenes	Medium	3 Questions
Biomolecules	Low/Medium	3-4 Questions
Alcohols, Phenols & Ethers	Low/Medium	2-3 Questions

STEP 3 — MOST REPEATED TOPICS (TOP 20)

1. **Nernst Equation** (3/3)
2. **First Order Kinetics Proof** (3/3)
3. **SN1 vs SN2 Reactivity** (3/3)
4. **Colligative / Van't Hoff** (3/3)
5. **Cannizzaro Reaction** (3/3)
6. **Kohlrausch Law** (3/3)
7. **CFT (Color / Splitting)** (3/3)
8. **Distinction Tests** (3/3)
9. **Lanthanoid Contraction** (2/3)
10. **Basicity of Amines** (3/3)
11. **d-Block Trends (E°)** (3/3)
12. **Glucose Reactions** (3/3)
13. **Arrhenius Eq (Ea)** (2/3)
14. **IUPAC (Coordination)** (3/3)
15. **VBT (Magnetic nature)** (2/3)
16. **Isomerism (Linkage/Optical)** (2/3)
17. **Prep of KMnO4/K2Cr2O7** (2/3)
18. **Phenol Acidity** (2/3)
19. **Electrolysis Products** (2/3)
20. **Protein Structure** (2/3)

STEP 4 — QUESTION PATTERN TYPES

- **Reasoning:** 30% (Account for following)
- **Numericals:** 25% (Solutions, Electro, Kinetics)
- **Direct Reactions/Conversions:** 20% (Organic)
- **MCQ & Assertion-Reason:** 15% (Section A)
- **Case-Based Studies:** 10% (Biomolecules/Electro)

STEP 5 — EXPECTED QUESTION AREAS

🔥 **MUST PREPARE:** One numerical each for: Nernst Eq, Integrated Rate Law (k=...), and Cryoscopy (i-factor). Name reactions: Cannizzaro, Aldol, Hoffmann Bromamide.

⚡ **HIGH CHANCE:** Glucose reacting with HI/HNO_3. Tollen's vs Iodoform distinction. Faraday's Law definition.

🎯 **BONUS:** Fuel Cell reactions, Arrhenius plot (ln k vs 1/T).

STEP 6 — EXPECTED PRACTICE QUESTIONS SET

Q1 (Kinetics): Show for first-order reaction that $t_{99\%} = 2 \times t_{90\%}$. (Medium)

Q2 (Electro): Calculate E_{cell} & $\Delta G°$ for: Mg(s) | Mg^{2+}(0.001M) || Cu^{2+}(0.0001M) | Cu(s). (Medium)

Q3 (Solutions): Calculate F.P. of 10g of $MgCl_2$ in 200g water (i=3). (Medium)

Q4 (Organic): Write prodotti for Cannizzaro reaction of Benzaldehyde. (Easy)

Q5 (Coordination): Why is $[Ni(CN)_4]^{2-}$ diamagnetic but $[Ni(Cl)_4]^{2-}$ paramagnetic? (Hard)

Q6 (Amines): Order of basicity in aqueous solution: $(CH_3)_2NH$, CH_3NH_2, $(CH_3)_3N$. (Tricky)

STEP 7 — LAST 7 DAYS QUICK REVISION

Essential Formulas:

- $E_{cell} = E° - (0.059/n) \log Q$
- $k = (2.303/t) \log([A]_0/[A])$
- $\Delta T_f = i \cdot K_f \cdot m$
- $\mu = \sqrt{[n(n+2)]}$ BM

Common Traps ⚠️:

Unit Error: Forgot to convert °C to Kelvin in Kinetics.

i-Factor: Forgot i=2 for NaCl or i=3 for $MgCl_2$ in numericals.

Sign Error: $\Delta G = -nFE$ (The minus sign is critical).

Amine order: Order is different in Aqueous vs Gas phase.

STEP 8 — HONEST REPORT

- **Syllabus Mismatch:** None. Alignment is perfect with 2024 Reduced Syllabus.
- **Excluded Parts:** Zero. All PDF content is relevant.
- **Confidence: Very High.** Patterns are consistent. Focus on Electrochemistry & Name Reactions for maximum returns.